(ce qui donne quatre systèmes de valeurs de $\lambda : \mu : \nu$), et puis

$$\Omega = af\,\frac{\nu - \mu}{\lambda} + bg\,\frac{\lambda - \nu}{\mu} + ch\,\frac{\mu - \lambda}{\nu}\ :$$

la surface devient une surface réglée, savoir, la *quadricuspidale* de M. de la Gournerie ; et, en supposant

$$(af)^{\frac{1}{3}} + (bg)^{\frac{1}{3}} + (ch)^{\frac{1}{3}} = 0,$$

et

$$\lambda : \mu : \nu \ = \ (af)^{\frac{1}{3}}\ :\ (bg)^{\frac{1}{3}}\ :\ (ch)^{\frac{1}{3}},$$

ce qui donne

$$\Omega \ = \ \left[(af)^{\frac{1}{3}} - (bg)^{\frac{1}{3}}\right]\left[(bg)^{\frac{1}{3}} - (ch)^{\frac{1}{3}}\right]\left[(ch)^{\frac{1}{3}} - (af)^{\frac{1}{3}}\right]\ :$$

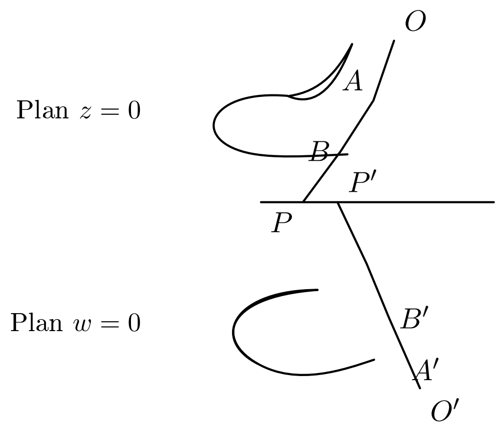
la surface devient développable.

Cambridge, 18 Octobre 1866.

Deuxième Mémoire. Notes pp. 279–283.

Note I. Sur la décomposition du lieu des génératrices en surfaces tétraédrales distinctes.

Il me semble qu'une de vos conclusions a besoin d'être modifiée. Ainsi la surface tétraédrale dérivée de deux courbes triangulaires à exposant $\frac{1}{m}$ (m étant un entier positif), laquelle, selon un de vos théorèmes, serait de l'ordre $2m^2$, paraît se décomposer en m surfaces chacune de l'ordre $2m$. Il y a pour cela une raison à *priori* ; en effet,



pour deux triangulaires de la forme en question, en employant votre construction, on peut établir une correspondance non-seulement entre les deux systèmes de points $A, B, \ldots$, et $A', B', \ldots$, mais aussi entre chaque point A et un seul point correspondant A', car l'équation de la première courbe étant de la forme

$$fx^{\frac{1}{m}} + gy^{\frac{1}{m}} + hz^{\frac{1}{m}} = 0,$$

on satisfait à cette équation en écrivant

$$x : y : z \;=\; a\,(\theta+\alpha)^m \;:\; b\,(\theta+\beta)^m \;:\; c\,(\theta+\gamma)^m,$$

où θ est un paramètre variable. De même, l'équation de la seconde courbe étant

$$f' x^{\frac{1}{m}} + g' y^{\frac{1}{m}} + k' w^{\frac{1}{m}} = 0,$$

on satisfait à cette équation en écrivant

$$x : y : w \;=\; a'\,(\theta'+\alpha')^m \;:\; b'\,(\theta'+\beta')^m \;:\; d'\,(\theta'+\delta')^m,$$

où θ' est aussi un paramètre variable.

Pour la droite OP, on a

$$\frac{x}{y} \;=\; \frac{a\,(\theta+\alpha)^m}{b\,(\theta+\beta)^m},$$

et pour la droite $O'P'$

$$\frac{x}{y} \;=\; \frac{a'\,(\theta'+\alpha')^m}{b'\,(\theta'+\beta')^m};$$

donc, la condition pour la correspondance des droites est

$$\frac{a\,(\theta+\alpha)^m}{b\,(\theta+\beta)^m} \;=\; \frac{a'\,(\theta'+\alpha')^m}{b'\,(\theta'+\beta')^m},$$

ce qui donne m valeurs différentes pour θ' en termes de θ. Mais chacune de ces valeurs est de la forme

$$\theta' \;=\; \frac{A\theta+B}{C\theta+D},$$

et, en ne faisant attention qu'à une seule valeur de θ', on a le point

$$x : y : z \;=\; a\,(\theta+\alpha)^m \;:\; b\,(\theta+\beta)^m \;:\; c\,(\theta+\gamma)^m,$$

qui correspond à un point unique

$$x : y : w \;=\; a'\,(\theta'+\alpha')^m \;:\; b'\,(\theta'+\beta')^m \;:\; d'\,(\theta'+\delta')^m.$$

Pour le cas de l'exposant $\frac{1}{3}$, on a, de cette manière, une surface de l'ordre 6. J'ai vérifié cela dans le cas particulier de la surface développable. Il est très-singulier (c'est M. Salmon qui m'a fait cette remarque) qu'en écrivant dans cette équation (x^2, y^2, z^2, w^2) au lieu de (x, y, z, w), on obtient l'équation d'une surface du douzième ordre, lieu des centres de courbure d'un ellipsoïde.

Cambridge, 15 Février 1866.

[433] p. 53

Note II. À l'occasion de l'ordre des surfaces tétraédrales.

Je crois que j'ai négligé de vous faire connaître un théorème assez général au sujet de l'ordre de ces surfaces. En considérant dans l'espace deux courbes (planes ou à double courbure) des ordres m et m' respectivement, et en supposant qu'il y ait entre les points de ces deux courbes une correspondance (a, a'), (c'est-à-dire qu'à un point donné de la courbe m correspondent a' points sur la courbe m', et à un point donné de la courbe m' correspondent a points sur la courbe m), alors la surface réglée que l'on obtient en unissant par des droites les points correspondants des courbes m et m' sera de l'ordre $ma' + m'a$.

Cambridge, 18 Octobre 1866.

Note III. Sur la surface complémentaire.

Je puis reconnaître, par mes propres formules, que, des pq^2 surfaces de l'ordre $2p^2q$, il n'y en a que pq qui passent par la troisième directrice. En effet, le rapport anharmonique k est donné en termes des paramètres de la troisième directrice, au moyen d'une équation qui contient la quantité irrationnelle $k^{\frac{p}{q}}$. En rationalisant cette équation, on obtient pour k une équation de l'ordre pq ; à chaque racine k_1 correspondent q surfaces, savoir celles qui appartiennent aux q valeurs de $k_1^{\frac{p}{q}}$; mais l'équation irrationnelle n'est satisfaite que par une seule valeur de $k_1^{\frac{p}{q}}$, à savoir la valeur de $k_1^{\frac{p}{q}}$ donnée par l'équation irrationnelle, en y substituant pour k, en tant que k y entre rationnellement, la valeur $k = k_1$. Donc, à chaque racine k_1 correspond une seule surface qui passe par la troisième directrice. La question à laquelle donne lieu cette circonstance paraît très-intéressante. La surface déterminée par les trois directrices est composée de pq surfaces chacune de l'ordre $2p^2q$, et d'une surface résiduale de l'ordre $2p^2(q^2 - q)$. Quelles sont la nature et les propriétés de cette surface résiduale ? Je serais bien aise de savoir si vous avez fait des recherches à ce sujet.

Cambridge, 29 Mars 1866.

[434]

On certain Skew Surfaces, otherwise Scrolls.

[From the Transactions of the Cambridge Philosophical Society, vol. XI. Part II. (1869), pp. 277–289. Read Nov. 11, 1867.]

THE investigations contained in the present Memoir were suggested to me by the Memoirs of M. De la Gournerie, presented by him to the Academy of Sciences in 1865 and 1866, published in extract in the *Comptes Rendus*, and reproduced in his work "Recherches sur les surfaces réglées tétraédrales symétriques, par Jules De la Gournerie, avec des notes par Arthur Cayley," 8vo. Paris, 1867. Although the results, or the greater part of them, agree with those in the work just referred to, the mode of treatment is different, and more general, the orders, &c. of the different scrolls being obtained by considerations founded on the theory of Correspondence; and I have thought it not improper to submit to geometers in this altered form the theory of the very interesting class of Scrolls for which they are indebted to M. De la Gournerie's researches.

Article Nos. 1 to 10. Geometrical Construction of a Class of Scrolls.

1. Consider any two curves (plane or of double curvature) U, U', of the orders m, m' respectively, and let the points of U have with those of U' an (α, α') correspondence; viz. let the points of the two curves be so related that to each point of U correspond α' points of U', and to each point of U' correspond α points of U; then the lines joining the corresponding points of U, U' form a scroll the order of which is $= m\alpha' + m'\alpha$.

2. In particular let U, U' be plane curves in the planes Π, Π' respectively; and let the correspondence between the points of the two curves be established as follows; viz. consider in the plane Π a curve Ω of the class μ, and in the plane Π' a curve Ω' of the class μ'; and (to avoid useless generality) let the tangents of these two curves Ω, Ω' have to each other a $(1, 1)$ correspondence; that is, to each tangent of Ω there corresponds a single tangent of Ω', and to each tangent of Ω' a single tangent of Ω (this assumes that the curves Ω, Ω' are rational transformations one of the other, and that they have consequently the same Deficiency). This being so, let the points of U which lie on any tangent of Ω and the points of U' which lie on the corresponding tangent of Ω' be taken to be corresponding points of U, U'. The correspondence is then $(\mu m', \mu' m)$: in fact, through a given point of U there pass μ tangents of Ω, and the corresponding μ tangents of Ω' meet U' in $\mu m'$ points, that is, to a given point of U correspond $\mu m'$ points of U'; and similarly to a given point of U' correspond $\mu' m$ points of U. And hence the order of the scroll formed by the lines joining the corresponding points of U, U' is $= (\mu + \mu')\, mm'$.

3. This conclusion may be otherwise established as follows; let K, K' be any two corresponding points of U, U', so that the scroll we are concerned with is that generated by the series of lines KK'; and let I denote the line of intersection of the planes Π, Π'. The line I meets the curve U in m points, and taking one of these points for a point K we may from this point draw μ tangents to the curve Ω, that is, the point in question is a point K in respect of μ different tangents of the curve Ω; to each of these tangents there corresponds a single tangent of Ω', and such tangent of Ω' meets the curve U' in m' points, that is, to the point K in question there correspond $\mu m'$ points K' and consequently $\mu m'$ lines KK' in the plane Π'; hence to each of the m points K on the line I there correspond $\mu m'$ lines KK' in the plane Π', and we have thus $\mu m m'$ generating lines in the plane Π'; there are in like manner $\mu' m m'$ generating lines in the plane Π.

Take K an arbitrary point on the curve U; there are $\mu m'$ corresponding points K', and consequently $\mu m'$ generating lines through K, that is, through each point of the curve U; or the curve U (which is of the order m) is a $\mu m'$-tuple line on the scroll; similarly the curve U' (which is of the order m') is a $\mu' m$-tuple line on the scroll.

The complete section of the scroll by the plane Π' consists of the curve U taken $\mu m'$ times (order $\mu m m'$) and of the $\mu' m m'$ generating lines in the plane Π'; that is, the order of the section is $= (\mu + \mu')\, mm'$; and we thus see that the order of the scroll is $= (\mu + \mu')\, mm'$. Of course in like manner the complete section of the scroll by the plane Π' consists of the curve U' taken $\mu' m$ times (order $\mu' m m'$) and of the $\mu m m'$ generating lines in the plane Π', the order of the section being thus $= (\mu + \mu')\, mm'$.

4. There are on the scroll certain singular tangent planes; viz. if we have two corresponding tangents of Ω, Ω' meeting the line I in the same point, then we have m points K and m' points K' all lying in the plane of the two tangents; and of course the mm' lines KK' will all lie in the plane of the two tangents, that is, the intersection of the scroll by the plane in question will be made up of the mm' lines, and of a curve of the order $(\mu + \mu' - 1)\, mm'$; and the plane in question is thus a singular tangent plane.

5. The number of these singular tangent planes is $= \mu + \mu'$; in fact, considering as corresponding points on the line I, the intersection of this line by any tangent of Ω and the intersection by the corresponding tangent of Ω', the correspondence is obviously (μ, μ'); viz. through a given point P considered as belonging to the first system there pass μ tangents of Ω, and corresponding thereto we have μ tangents of Ω' each intersecting I in a single point P'; that is, to a given point P correspond μ points P'; and similarly to a given point P' correspond μ' points P. And this being so, the number of united points, that is, points of I through which pass corresponding tangents of Ω, Ω', is $= \mu + \mu'$.

6. In particular the curves Ω, Ω' may reduce themselves each to a point: the tangents to the two curves are here the lines passing through the points Ω, Ω' respectively: and the condition for the $(1, 1)$ correspondence of the two tangents is that the pencils of lines shall be homographically related; or, what is the same thing, that these two pencils shall determine on the line I two ranges which are homographically related; the entire construction is then as follows:

Given in the plane Π a curve U and a point Ω, and in the plane Π' a curve U' and a point Ω'; and taking in the plane Π a pencil of lines through Ω, and in the plane Π' a pencil of lines through Ω', in such wise that the two pencils correspond homographically; then if a line of the first pencil meets the curve U in the m points K, and the corresponding line of the second pencil meets the curve U' in the m' points K', the scroll in question is that generated by the mm' lines KK'.

7. By what precedes, the scroll is of the order $2mm'$; the curve U is a m'-tuple line, and the complete section by the plane Π is made up of this curve taken m' times and of mm' generating lines; similarly the curve U' is a m-tuple line, and the complete section by the plane Π' is made up of this curve taken m times and of mm' generating lines; there are two singular tangent planes such that the section by each of them is made up of mm' generating lines and of a curve of the order mm'; the planes in question are obviously those through the line $\Omega\Omega'$ and the coincident points of the two ranges on the line I, say the points A, B respectively.

8. The foregoing results will be modified in special cases. Suppose, for instance, that the curve U passes ω times, α times, and β times through the points Ω, A, B respectively, and that the curve U' passes ω' times, α' times, and β' times through the points Ω', A, B respectively. Then to each point on the curve U there correspond the $m' - \omega'$ intersections (other than the point Ω') on a line through Ω', so that U' is a $(m' - \omega')$-tuple line on the surface. The curve U' meets the line I in m' points; and corresponding to each of them we have a line through Ω meeting the curve U in $(m - \omega)$ points, exclusive of the point Ω; this would give $m'(m - \omega)$ generating lines in the plane Π; but among the m' points are included the point A α' times, and the point B β' times; the $(m - \omega)$ points corresponding to A include the point A α times, and we have thus the point A corresponding to itself $\alpha\alpha'$ times, and giving a reduction $= \alpha\alpha'$ in the number $m'(m - \omega)$ of generating lines; similarly the $m - \omega$ points corresponding to B

include the point B β times, and we have thus the point B corresponding to itself $\beta\beta'$ times and giving a reduction $= \beta\beta'$ in the number $m'(m-\omega)$ of generating lines; the number of generating lines in the plane Π is thus $= m'(m-\omega) - \alpha\alpha' - \beta\beta'$. The complete section by the plane Π is made up of the curve U $(m-\omega')$ times (order $m(m'-\omega')$) and of the $m'(m-\omega) - \alpha\alpha' - \beta\beta'$ generating lines; the order of the section, and consequently also the order of the scroll, is thus $= 2mm' - m\omega' - m'\omega - \alpha\alpha' - \beta\beta'$. It is clear that in like manner the curve U' is a $(m-\omega)$-tuple line on the surface, and that the complete section by the plane Π' is made up of this curve taken $(m-\omega)$ times, order $m'(m-\omega)$, and of $m(m'-\omega') - \alpha\alpha' - \beta\beta'$ generating lines.

9. The section by the tangent plane through A is made up of $(m-\omega)(m'-\omega') - \alpha\alpha'$ generating lines (viz. these are, the line $\Omega' A$ $\alpha'(m-\omega-\alpha)$ times, the line ΩA $\alpha(m'-\omega'-\alpha')$ times, and $(m-\omega-\alpha)(m'-\omega'-\alpha')$ other generating lines) and of a curve of the order $mm'-\omega\omega'-\beta\beta'$; similarly the section by the tangent plane through B is made up of $(m-\omega)(m'-\omega') - \beta\beta'$ generating lines (viz. these are, the line ΩB $\beta'(m-\omega-\beta)$ times, the line $\Omega' B$ $\beta(m'-\omega'-\beta')$ times, and $(m-\omega-\beta)(m'-\omega'-\beta')$ other generating lines), and of a curve of the order $mm'-\omega\omega'-\alpha\alpha'$.

10. A very interesting case is when (m, m' being each even) we have

$$\omega = \alpha = \beta = \tfrac{1}{2}m, \qquad \omega' = \alpha' = \beta' = \tfrac{1}{2}m'.$$

Here the curve U is a $\frac{1}{2}m'$-tuple line on the scroll, and the complete section by the plane Π is this curve taken $\frac{1}{2}m'$ times; the order of the section, and therefore of the scroll, is thus $= \frac{1}{2}mm'$: of course in like manner the curve U' is a $\frac{1}{2}m$-tuple line on the scroll, and the complete section by this plane is the curve U' taken $\frac{1}{2}m$ times; the section by each of the planes $\Omega\Omega' A$, $\Omega\Omega' B$ is a curve of the order $\frac{1}{2}mm'$, the planes in question being in the present case no longer singular tangent planes, or even tangent planes at all, of the scroll.

Article Nos. 11 to 14. Analytical Theory.

11. It will be convenient to denote by D, C respectively the points heretofore called Ω, Ω' respectively: this being so, we have a tetrahedron $ABCD$, of which the faces ABD, ABC are the planes heretofore called Π, Π' respectively, and the other two faces CDA, CDB are the singular tangent planes $\Omega\Omega' A$, $\Omega\Omega' B$ respectively. And then, taking

$$x = 0, \quad y = 0, \quad z = 0, \quad w = 0$$

for the equations of the faces BCD, CDA, DAB, ABC of the tetrahedron, we may write for the equations of the curve U, $z = 0$, $f_1(x, y, w) = 0$, for those of the curve U', $w = 0$, $f_2(x, y, z) = 0$; and take the homographic ranges on the line I ($z = 0$, $w = 0$) to be given as the intersections of this line with the pencils of planes $x - \theta y = 0$, $x - k\theta y = 0$ respectively (θ a variable parameter, k a constant). The points K are therefore given by

$$x - \theta y = 0, \qquad z = 0, \qquad f_1(x, y, w) = 0,$$

the points K' by

$$x - k\theta y = 0, \qquad w = 0, \qquad f_2(x, y, z) = 0;$$

and then the lines KK' belonging to the different values of the parameter θ generate the scroll.

12. Or, what is the same thing, taking (X, Y, Z, W) as the coordinates of K, (X', Y', Z', W') as the coordinates of K', we have

$$\begin{array}{lll} X - \theta Y = 0, & Z = 0, & f_1(X, Y, W) = 0, \\ X' - k\theta Y' = 0, & W' = 0, & f_2(X', Y', Z') = 0, \end{array}$$

and then the equations of the line KK' are

$$\begin{vmatrix} x & y & z & w \\ X & Y & 0 & W \\ X' & Y' & Z' & 0 \end{vmatrix} = 0;$$

or, as these may be written,

$$\begin{aligned} -WZ'y + W'Y'z + YZ'w &= 0, \\ WZ'x \quad + W'X'z - XZ'w &= 0, \\ -W'Yx + WX'y + (XY' - X'Y)\,w &= 0, \\ -YZ'x + XZ'y - (XY' - X'Y)\,z &= 0, \end{aligned}$$

equivalent of course to two equations. The elimination of X, Y, W, X', Y', Z', θ from all the equations gives the equation of the scroll.

13. Substituting the values $X = \theta Y$, $X' = k\theta Y'$, we have

$$f_1(\theta Y, Y, W) = 0, \qquad f_2(k\theta Y', Y', Z') = 0,$$

and

$$\begin{aligned} -WZ'y + WY'z + YZ'w &= 0, \\ WZ'x - k\theta WY'z - \theta YZ'w &= 0, \\ -WY'x + k\theta WY'y + \theta(1-k)\,YY'w &= 0, \\ -YZ'x + \theta YZ'y - \theta(1-k)\,YY'z &= 0; \end{aligned}$$

or, what is the same thing, writing $\dfrac{W}{Y} = \omega$ and $\dfrac{Z'}{Y'} = \zeta$, we have

$$f_1(\theta, 1, \omega) = 0, \qquad f_2(k\theta, 1, \zeta) = 0,$$

$$\begin{aligned} \omega\zeta\,x - \omega\zeta\,y \quad + \zeta\,\omega &= 0, \\ \omega\zeta\,x - k\theta\,\omega\zeta\,z - \theta\zeta\,\omega &= 0, \\ -\,\omega\,x + k\theta\,\omega\,y \quad + \theta(1-k)\,\omega &= 0, \\ -\,\zeta\,x + \theta\zeta\,y - \theta(1-k)\,z &= 0. \end{aligned}$$

Recollecting that the last four equations are equivalent to two equations only, and substituting for ω, ζ their values in terms of θ, we have in effect two equations, which by the elimination of θ lead to a relation in (x, y, z, w), the equation of the scroll.

14. We may find the sections of the scroll by the planes $x = 0$, $y = 0$ respectively. Writing first $x = 0$, we have

$$\omega y = \frac{k-1}{k}\,w, \qquad \zeta y = -(k-1)\,z.$$

Hence taking the other two equations in the form

$$f_1\Big(\theta y, y, \frac{k-1}{k}\,w\Big) = 0, \qquad f_2\Big(u, \frac{y}{k}, \frac{-(k-1)z}{k}\Big) = 0,$$

and putting $\theta y = u$, we have

$$f_1\Big(u, y, \frac{k-1}{k}\,w\Big) = 0, \qquad f_2\Big(u, \frac{y}{k}, \frac{-(k-1)z}{k}\Big) = 0,$$

from which eliminating u we obtain an equation $f_3(y,z,w)=0$, the equation of the section by the plane $x=0$.

Similarly, writing $y=0$, we have

$$\frac{\omega x}{\theta}=-(k-1)\,w,\qquad \frac{\zeta x}{\theta}=(k-1)\,z,$$

whence taking the equations in the form

$$f_1\Big(x,\frac{x}{\theta},\frac{-(k-1)w}{\theta}\Big)=0,\qquad f_2\Big(kx,v,\frac{(k-1)z}{\theta}\Big)=0,$$

and writing $\dfrac{x}{\theta}=v$, we have

$$f_1(x,v,-(k-1)\,w)=0,\qquad f_2(kx,v,(k-1)\,z)=0,$$

from which, eliminating v, we obtain an equation $f_4(x,z,w)=0$, the equation of the section by the plane $y=0$.

Article Nos. 15 to 29. The Curves U, U' are henceforward "triangular" curves.

15. Let $r=\pm\dfrac{p}{q}$, where p, q are positive integers prime to each other, and let the given sections be

$$\begin{aligned} z=0,\quad & Ax^r+By^r \qquad\quad +Dw^r=0,\\ w=0,\quad & A'x^r+B'y^r+C'z^r \qquad\quad =0,\end{aligned}$$

where it is to be observed that r being $=+\dfrac{p}{q}$, the two given sections are of the order pq, the order of the scroll is $=2p^2q^2$; each of the given sections is a pq-tuple line on the scroll, and the plane thereof meets the scroll in the section taken pq times, and in the pq generating lines: but r being $=-\dfrac{p}{q}$, the two given sections are each of the order $2pq$, with three pq-tuple points ($\omega=\alpha=\beta=pq$, $\omega'=\alpha'=\beta'=pq$), and thence the order of the scroll is $\frac{1}{2}(2pq)^2=2p^2q^2$; each of the sections is a pq-tuple line on the scroll, and the plane meets the scroll only in the section taken pq times. But in either case, if q be >1, that is, if r be fractional, it will presently appear that the scroll of the order $2p^2q^2$ breaks up into q scrolls each of the order $2p^2q$.

16. To find the section by the plane $x=0$, we have

$$Au^r+By^r \qquad\quad +D\Big(\frac{k-1}{k}\,w\Big)^r=0,$$

$$A'(ku)^r+B'y^r+C'\Big(\frac{-(k-1)z}{k}\Big)^r=0,$$

and eliminating u we obtain

$$(AB'-A'Bk^r)\,y^r+AC'\Big(\frac{-(k-1)}{k}\Big)^r z^r-A'D\Big(\frac{k-1}{k}\Big)^r w^r=0;$$

writing $\left(\dfrac{k-1}{k}\right)^r=(-)^r\left(\dfrac{1-k}{k}\right)^r$, this is

$$\frac{AB'-A'Bk^r}{(1-k)^r}\,y^r-(-)^r\,AC'\,z^r+A'D\,w^r=0;$$

or, what is the same thing, it is

$$\frac{AB' - A'B\,k^r}{(1-k)^r}\,y^r - (-)^r\,AC'\,z^r + A'D\,w^r = 0.$$

And in regard to this and the other equations which contain $(-)^r$, it is to be observed that r being integral we have $(-)^r = (-)^r$, and that r being fractional, every value of $(-)^r$ is also a value of $(-)^r$; so that we may in every case write $(-)^r$ in place of $(-)^{-r}$.

Similarly for the section by the plane $y = 0$, we have

$$\begin{aligned} &Ax^r + Bv^r &&+ D\,(-(k-1)w)^r = 0,\\ &A'\,(kx)^r + B'v^r + C'\,((k-1)z)^r &&= 0, \end{aligned}$$

and eliminating v, we have

$$(AB' - A'Bk^r)\,x^r - BC'\,(k-1)^r\,z^r + BD\,(-(k-1))^r\,w^r = 0;$$

or, what is the same thing,

$$\frac{AB' - A'Bk^r}{(1-k)^r}\,x^r - (-)^r\,BC'\,z^r + B'D\,w^r = 0.$$

17. The four sections thus are

$$\begin{aligned} x &= 0, && \frac{AB' - A'Bk^r}{(1-k)^r}\,y^r - (-)^r\,AC'\,z^r + A'D\,w^r = 0,\\ y &= 0, && \frac{AB' - A'Bk^r}{(1-k)^r}\,x^r - (-)^r\,BC'\,z^r + B'D\,w^r = 0,\\ z &= 0, && Ax^r + By^r \qquad + Dw^r = 0,\\ w &= 0, && A'x^r + B'y^r + C'z^r \qquad = 0. \end{aligned}$$

It will be convenient to speak of these four curves as directrices of the scroll.

18. Suppose for a moment that r is integral; as either of the given equations may be multiplied by a constant, we may assume that $D = -(-)^r\,C'$; substituting this value and dividing the first and second equations each by C', we have

$$\begin{aligned} x &= 0, && \frac{AB' - A'Bk^r}{(1-k)^r\,C'}\,y^r - (-)^r\,Az^r - A'w^r = 0,\\ y &= 0, && \frac{AB' - A'Bk^r}{(1-k)^r\,C'}\,x^r - (-)^r\,Bz^r - B'w^r = 0,\\ z &= 0, && Ax^r + By^r \qquad - (-)^r\,C'w^r = 0,\\ w &= 0, && A'x^r + B'y^r + C'z^r \qquad = 0, \end{aligned}$$

so that the diagonally opposite coefficients differ only by the factor $-(-)^r$; viz. the matrix is symmetrical or skew symmetrical according as r is odd or even.

19. If r be fractional, it is to be observed that, although the three symbols $(-)^r$ and the two symbols $(1-k)^r$ which enter into the first and second equations of No. 17, do not in the first instance represent of necessity the same values of $(-)^r$ and $(1-k)^r$ respectively, yet there is no loss of generality in assuming that they do so — the irrational equations are mere symbols for the rational equations to which they respectively give rise — and the irrationalities $(-)^r$ and $(1-k)^r$ will on the rationalisation of the equations disappear along with the irrationalities of x^r, y^r, z^r, w^r to which they are attached. But the case is otherwise with the irrationality k^r

involved in the expression $AB' - A'Bk^r$; writing as before $r = \pm\frac{p}{q}$ (p and q positive integers prime to each other), the symbol k^r has q different values; and there is not in the first instance any relation between the k^r of the first equation and the k^r of the second equation: for each of these equations the rationalised equation (that is, the equation rationalised in regard to the coordinates) will contain the irrationality k^r, and will thus for each of the q values of k^r represent a distinct curve. The given equations (viz. the first and second equations) represent each of them a single curve of the order pq or $2pq$, according as r is positive or negative; the first and second equations represent each of them q such curves.

20. Hence, starting from the two given curves in the planes $z = 0$ and $w = 0$, respectively, and with a given value of k, the section of the scroll by the plane $y = 0$ is made up of q curves, viz. the curves obtained from the second equation of No. 17, by assigning to the radical k^r each of its q different values; the scroll consequently breaks up into q different scrolls, viz. the lines passing through the two given curves, and any one of the q curves in the plane $y = 0$, constitute a distinct scroll. The lines in question meet the plane $x = 0$, not indifferently in any one of the q curves in that plane, but in a certain one of these curves, viz. in that curve for which the radical k^r has the same value as for the curve in question in the plane $y = 0$. Hence we may in the first and second equations regard the radicals k^r as having the same meaning, and the system of four equations in effect breaks up into q systems, viz. the systems obtained by giving to the radical k^r its q different values; each of these systems gives a scroll, and the scroll derived from the two given curves with a given value of k is made up of these q scrolls. And hence, attaching a unique value to each of the symbols $(-)^r$, $(1-k)^r$ and k^r, we may, as before, write $D = -(-)^r\,C'$, and so reduce the original equations as in the case r integral, to the form No. 18, in which the diagonally opposite coefficients differ only by the factor $-(-)^r$.

21. Let the two given equations be taken to be

$$\begin{aligned} z &= 0, \quad bx^r + yay^r \qquad\quad + hw^r &&= 0,\\ w &= 0, \quad -(-)^r fx^r - (-)^r gy^r + (-)^r cz^r &&= 0; \end{aligned}$$

we have then

$$\frac{AB' - A'Bk^r}{(1-k)^r\,C'} = \frac{(-)^r\,bg + afk^r}{(-)^r\,c\,(1-k)^r},$$

or, putting this $= -(-)^r\,c$, that is, writing

$$af\,k^r + bg\,(-)^r + ch\,(1-k)^r = 0,$$

the four equations become

$$\begin{aligned} x &= 0, & + cy^r - (-)^r\,bz^r + fw^r &= 0,\\ y &= 0, & -(-)^r\,cx^r \qquad + az^r + gw^r &= 0,\\ z &= 0, & bx^r - (-)^r\,ay^r \qquad + hw^r &= 0,\\ w &= 0, & -(-)^r\,fx^r - (-)^r\,gy^r - (-)^r\,hz^r &= 0, \end{aligned}$$

where c being considered as given, k is determined as mentioned above; or, what is the same thing, $k : -1 : 1-k = \lambda : \mu : \nu$, we have $\lambda : \mu : \nu$, and thence k, determined by the equations

$$\lambda + \mu + \nu = 0, \qquad af\,\lambda^r + bg\,\mu^r + ch\,\nu^r = 0.$$

22. Consider for a moment λ, μ, ν, as the coordinates of a point in a plane, then $\big(r = \pm\frac{p}{q}$ as before$\big)$, the equation $af\,\lambda^r + bg\,\mu^r + ch\,\nu^r = 0$, is that of a curve of the order pq or $2pq$, according as r is positive or negative: and this curve is met by the line $\lambda + \mu + \nu = 0$, in pq or $2pq$ points, that is, k has this number pq or $2pq$, of values: but to each of these values of k there corresponds (not q values but) only a single value of k^r, viz. that value for which $af\,k^r + bg\,(-)^r + ch\,(1-k)^r = 0$; that is, starting from the two directrices in the planes $z = 0$, $w = 0$, respectively, and a given third directrix in the plane $y = 0$ (or in the plane $x = 0$), we may by means of each of the pq or $2pq$ values of k construct a scroll passing through the three directrices, and which will also pass through the fourth directrix in the plane $x = 0$ (or in the plane $y = 0$), but such scroll is only one (not each) of the q scrolls which can be constructed from the two given sections in the planes $z = 0$, $w = 0$, respectively, and from the assumed value of k. It has been mentioned that whether r is $= +\dfrac{p}{q}$, or $= -\dfrac{p}{q}$, the total scroll constructed from the two given directrices in the planes $z = 0$, $w = 0$, and from a given value of k is of the order $2p^2q^2$, and that such scroll breaks up into q distinct scrolls, hence the order of each of the distinct scrolls is $= 2p^2q$. Whence, starting with the given directrices in the planes $z = 0$, $w = 0$, and a given third directrix in the plane $y = 0$ (or in the plane $x = 0$), we have pq or $2pq$ scrolls each of the order $2p^2q$, and passing through these three directrices, and through the given fourth directrix in the plane $x = 0$ (or in the plane $y = 0$).

23. It is to be observed that when q is > 1, then considering the three directrices as given, the pq or $2pq$ scrolls each of the order $2p^2q$, do not make up the total scroll generated by the lines which pass through the three given directrices. I call to mind that for three given directrices the orders of which are m, n, p, respectively, and which meet, the second and third, the third and first, and the first and second, in α points, β points, and γ points respectively, the order of the scroll generated by the lines which meet the three directrices is $= 2mnp - \alpha m - \beta n - \gamma p$. Suppose first, that $r = +\dfrac{p}{q}$, then the directrices are each of the order pq, and they do not any two of them meet; the order of the scroll is $= 2p^3q^3$. Suppose secondly, $r = -\dfrac{p}{q}$, then the directrices are each of the order $2pq$, but each two of them have in common two pq-tuple points counting as $2p^2q^2$ intersections; the order of the scroll is thus $(16 - 2\cdot 4)\,p^3q^3 = 4p^3q^3$. In the first case the lines which meet the three directrices generate a residuary scroll of the order $2p^3(q^3 - q^2)$, and the pq scrolls each of the order $2p^2q$; in the second case they generate a residuary scroll of the order $4p^3(q^3 - q^2)$, and the $2pq$ scrolls each of the order $2p^2q$.

24. In the case $r = +\dfrac{p}{q}$, by way of illustration of the origin of the pq scrolls each of the order $2p^2q$, I consider the particular case $p = 1$, that is, $r = \dfrac{1}{q}$, the reciprocal of a positive integer q, and where it is to be shown that we have q scrolls each of the order $2q$. The given directrices are here

$$\begin{aligned} z = 0, \quad & Ax^{\frac{1}{q}} + By^{\frac{1}{q}} \phantom{{}+C'z^{\frac{1}{q}}} + Dw^{\frac{1}{q}} = 0, \\ w = 0, \quad & A'x^{\frac{1}{q}} + B'y^{\frac{1}{q}} + C'z^{\frac{1}{q}} \phantom{{}+Dw^{\frac{1}{q}}} = 0, \end{aligned}$$

each of them a unicursal curve; we may in fact satisfy the two equations respectively, by writing in the first of them

$$x : y : w \;=\; a\,(\phi + \alpha)^q \;:\; b\,(\phi + \beta)^q \;:\; d\,(\phi + \delta)^q;$$

and in the second

$$x : y : z \;=\; a'\,(\phi' + \alpha')^q \;:\; b'\,(\phi' + \beta')^q \;:\; c'\,(\phi' + \gamma')^q,$$

where a, b, d, α, β, δ, a', b', c', α', β', γ' are properly determined constants, ϕ, ϕ' are variable parameters. It follows that, considering the points K, K' which are the intersections of the first curve by the line $x - \theta y = 0$, and of the second curve by the corresponding line $x - k\theta y = 0$, we have not only a correspondence of q points K with q points K', but we may establish, and that in q different manners, a correspondence between single points K and K'. For, substituting the foregoing values of $x : y$ in the equations $x - \theta y = 0$ and $x - k\theta y = 0$ respectively, we have

$$\theta = \frac{a\,(\phi+\alpha)^q}{b\,(\phi+\beta)^q}, \qquad k\theta = \frac{a'\,(\phi'+\alpha')^q}{b'\,(\phi'+\beta')^q},$$

and thence

$$\frac{a\,(\phi+\alpha)^q}{b\,(\phi+\beta)^q} = \frac{1}{k}\cdot\frac{a'\,(\phi'+\alpha')^q}{b'\,(\phi'+\beta')^q};$$

so that, extracting the qth root of each side, we have, in q different ways corresponding to the q values of the radical $\left(\frac{1}{k}\cdot\frac{ba'}{ab'}\right)^{\frac{1}{q}}$, a relation of the form $\phi' = \frac{\Lambda\phi + M}{N\phi + R}$; and considering ϕ' as having this value, the points K, K' as given by the equations

$$\begin{aligned} z = 0, \quad x : y : w &= a\,(\phi+\alpha)^q : b\,(\phi+\beta)^q : d\,(\phi+\delta)^q, \\ w = 0, \quad x : y : z &= a'\,(\phi'+\alpha')^q : b'\,(\phi'+\beta')^q : c'\,(\phi'+\gamma')^q, \end{aligned}$$

respectively, correspond as single points to each other. We have thus in q different ways a series of corresponding points K, K', and consequently q series of lines KK' each of them generating a scroll which (as the order of the scroll generated by all the q series is $= 2q^2$), must be each of them of the order $2q$; and the decomposition in question is thus explained.

25. In the scroll of the order $2q^2$, each directrix is a q-tuple line, and the complete section by the plane of the directrix is made up of the directrix q times (order q^2), and of q q-fold generating lines, in fact, of q q-fold generating lines: to show that this is so, consider the directrix in the plane $z = 0$, viz. the equation of this is $Ax^{\frac{1}{q}} + By^{\frac{1}{q}} + Dw^{\frac{1}{q}} = 0$. Writing herein $w = 0$, we have $Ax^{\frac{1}{q}} + By^{\frac{1}{q}} = 0$, that is, $A^q x - (-)^q B^q y = 0$; it is clear that the rationalised equation must reduce itself to $[A^q x - (-)^q B^q y]^q = 0$, and that the line $w = 0$, is thus a tangent of q-pointic intersection at the point $w = 0$, $A^q x - (-)^q B^q y = 0$. Taking K at this point we have, in each of the scrolls of the order $2q$, q coincident positions of K', that is, a q-fold line KK' in the plane $w = 0$; and the like for the plane $z = 0$, so that the total section by the plane $z = 0$ is made up of the directrix q times and of q q-fold generating lines; and it follows that for each of the scrolls of the order $2q$, the section by the plane $z = 0$ is made up of the directrix once, and of a q-fold generating line.

26. It is easy to see that in the general case $r = +\frac{p}{q}$, the like conclusion holds for the scroll of the order $2p^2q^2$, the section by the plane of the directrix consists of the directrix pq times (order p^2q^2), and of p^2q q-fold generating lines; whence for each of the q component scrolls of the order $2p^2q$, the section is made up of the directrix p times (that is, the directrix is a p-tuple line on the scroll) and of p^2 q-fold generating lines.

27. In the case $r = -\frac{p}{q}$, where the order of the directrix is $= 2pq$, then in the scroll of the order $2p^2q^2$, the directrix is a pq-tuple line on the scroll, and taken pq times it constitutes the complete section by the plane of the directrix; whence in each of the q component scrolls of the order $2p^2q$, the directrix is a p-tuple line; and taken p times it constitutes the complete section by the plane of the directrix.

28. It is convenient to exhibit the foregoing results in a tabular form as follows:

$r = +\frac{p}{q}$. Each directrix is of order pq.	$r = -\frac{p}{q}$. Each directrix is of order $2pq$, with three pq-tuple points.
Scroll belonging to two directrices, and a given value of k, is of the order	
$2p^2q^2$, breaking up into q scrolls each of order $2p^2q$, each which scroll of the order $2p^2q$ has each directrix for a p-tuple line and has besides p^2 q-fold generating lines in the plane of the directrix.	$2p^2q^2$, breaking up into q scrolls each of the order $2p^2q$, each which scroll of the order $2p^2q$ has each directrix for a p-tuple line, and consequently no generating line in the plane of the directrix.
Considering two directrices and a given third directrix,	
k has pq values.	k has $2pq$ values.
Total scroll for the three directrices is made up of	
pq scrolls each of order $2p^2q$ (viz. one for each value of k), and residuary scroll of order $2p^3(q^3 - q^2)$.	$2pq$ scrolls each of order $2p^2q$ (viz. one for each value of k), and residuary scroll of order $4p^3(q^3 - q^2)$.

29. The following are noticeable cases; $r = 1$ gives the hyperboloid as derived from three directrix lines; $r = -1$ the hyperboloid as derived from three plane sections thereof; $r = 2$, an octic surface, M. De la Gournerie's Quadrispinal; $r = -2$, an octic surface, his Quadricuspidal; $r = \frac{1}{3}$, a sextic surface which (as remarked by Dr Salmon), on writing therein (x^2, y^2, z^2, w^2), in place of (x, y, z, w), is converted into a surface of the twelfth order, locus of the centres of curvature of an ellipsoid.

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On the Six Coordinates of a Line.

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THE notion of the six coordinates of a line was, so far as I am aware, first established in my paper "On a new analytical representation of Curves in Space," *Quart. Math. Jour.* t. III. (1860), pp. 225–236, [**284**]; see p. 226, where writing p, q, r, s, t, u for the six determinants of the matrix $\left(\begin{smallmatrix} x_1 & y_1 & z_1 & w_1 \\ x_2 & y_2 & z_2 & w_2 \end{smallmatrix}\right)$, I remark that these values give identically $ps + qt + ru = 0$; and I consider a cone as represented by a homogeneous equation $V = 0$ between the six coordinates (p, q, r, s, t, u); and many of the investigations of the present memoir, in which these coordinates are employed, have been in my possession for some years past. But these coordinates presented themselves independently to Prof. Plücker, and the theory of them is set forth in his most interesting and valuable memoir, "On a new Geometry of Space," *Phil. Trans.* t. CLV. (1865), pp. 725–791; the course of development there given to the theory is however altogether different from that in the present memoir. They have also more recently been made use of in a paper by Herr Lüroth, "Zur Theorie der windschiefen Flächen," *Crelle*, t. LXII. (1867), pp. 130–152.

I have in the present memoir applied these coordinates to the question of the Involution of six lines; the notion of this relation of six lines is due to Prof. Sylvester, to whom it presented itself in the year 1861, in connexion with a theorem in the *Lehrbuch der Statik*, by Möbius (Leipzig, 1837), that if four forces acting on a solid body are in equilibrium the lines along which the forces act are the generating lines of a hyperboloid. Prof. Sylvester was thereby led to consider six lines such that (regarding them as lines in a solid body) there exist along them forces which are in equilibrium; and he thence obtained, by the statical considerations reproduced in the present memoir, the construction (when five of the lines are given) of a sixth line to pass through a given point or to be situate in a given plane.

Article, Nos. 1 to 8. The Six Coordinates of a Line; definition and general notions.

1. Using any quadriplanar coordinates (x, y, z, w) whatever, consider a line; on the line two points the coordinates of which are $(\alpha, \beta, \gamma, \delta)$ and $(\alpha', \beta', \gamma', \delta')$ respectively; and through the line two planes, the equations whereof are $(A, B, C, D \between x, y, z, w) = 0$, and $(A', B', C', D' \between x, y, z, w) = 0$ respectively; we have

$$\begin{aligned}(A, B, C, D \between \alpha, \beta, \gamma, \delta) &= 0,\\ (A, B, C, D \between \alpha', \beta', \gamma', \delta') &= 0,\\ (A', B', C', D' \between \alpha, \beta, \gamma, \delta) &= 0,\\ (A', B', C', D' \between \alpha', \beta', \gamma', \delta') &= 0.\end{aligned}$$

2. From the first and second equations, eliminating successively A, B, C, D, we find

$$\begin{vmatrix} 0 & \alpha\beta' - \alpha'\beta & -(\gamma\alpha' - \gamma'\alpha) & \alpha\delta' - \alpha'\delta \\ -(\alpha\beta' - \alpha'\beta) & 0 & \beta\gamma' - \beta'\gamma & \beta\delta' - \beta'\delta \\ \gamma\alpha' - \gamma'\alpha & -(\beta\gamma' - \beta'\gamma) & 0 & \gamma\delta' - \gamma'\delta \\ -(\alpha\delta' - \alpha'\delta) & -(\beta\delta' - \beta'\delta) & -(\gamma\delta' - \gamma'\delta) & 0 \end{vmatrix} (A, B, C, D) = 0,$$

and from the third and fourth equations we find the like system with (A', B', C', D') in place of (A, B, C, D). Comparing the corresponding equations of the two systems, we find an equality of ratios, as will presently be mentioned.

3. From the first and third equations, eliminating successively α, β, γ, δ, we find

$$\begin{vmatrix} 0 & AB'-A'B & -(CA'-C'A) & AD'-A'D \\ -(AB'-A'B) & 0 & BC'-B'C & BD'-B'D \\ CA'-C'A & -(BC'-B'C) & 0 & CD'-C'D \\ -(AD'-A'D) & -(BD'-B'D) & -(CD'-C'D) & 0 \end{vmatrix}(\alpha,\beta,\gamma,\delta)=0,$$

and from the second and fourth equations we find the like system with $(\alpha',\beta',\gamma',\delta')$ in place of $(\alpha,\beta,\gamma,\delta)$; comparing the corresponding equations of the two systems, we find the same equality of ratios as before, viz.

4. This is

$$\begin{aligned}\beta\gamma'-\beta'\gamma:\gamma\alpha'-\gamma'\alpha:\alpha\beta'-\alpha'\beta:\alpha\delta'-\alpha'\delta:\beta\delta'-\beta'\delta:\gamma\delta'-\gamma'\delta \\ = AD'-A'D:BD'-B'D:CD'-C'D:BC'-B'C:CA'-C'A:AB'-A'B,\end{aligned}$$

and putting each of these two equal sets of ratios

$$=a:b:c:f:g:h,$$

then the quantities (a,b,c,f,g,h), which it is easy to see satisfy the condition

$$af+bg+ch=0,$$

are said to be the 'six coordinates' of the line: as only the ratios of the six quantities are material, and as the last-mentioned equation establishes a single relation between these ratios, the system of the six coordinates contain four arbitrary ratios or parameters, for the determination of the particular line.

5. A line is thus determined by its six coordinates (a,b,c,f,g,h), which are such that $af+bg+ch=0$; and conversely any six quantities (a,b,c,f,g,h) satisfying this relation may be taken to be the six coordinates of a line.

6. It is proper to show that the ratios $a:b:c:f:g:h$ are independent of the particular two points on the line, or two planes through the line, used for their determination. In fact, if instead of the points

$$\alpha,\beta,\gamma,\delta, \qquad \alpha',\beta',\gamma',\delta',$$

we have any other two points on the line, say the points

$$\lambda\alpha+\mu\alpha',\ \ \lambda\beta+\mu\beta',\ \ \lambda\gamma+\mu\gamma',\ \ \lambda\delta+\mu\delta',$$

$$\nu\alpha+\rho\alpha',\ \ \nu\beta+\rho\beta',\ \ \nu\gamma+\rho\gamma',\ \ \nu\delta+\rho\delta',$$

then the six determinants have their original values each multiplied by $\lambda\rho-\mu\nu$; and the ratios are unaltered.

And the like is the case, if instead of the planes

$$A,B,C,D, \qquad A',B',C',D',$$

we have any other two planes through the line, say the planes

$$\lambda A+\mu A',\ \ \lambda B+\mu B',\ \ \lambda C+\mu C',\ \ \lambda D+\mu D',$$

$$\nu A+\rho A',\ \ \nu B+\rho B',\ \ \nu C+\rho C',\ \ \nu D+\rho D';$$

the determinants have their original values each multiplied by $\lambda\rho-\mu\nu$; and the ratios are unaltered.

7. It may be remarked, that the theory of the six coordinates considered as derived from the two points $(\alpha, \beta, \gamma, \delta)$, $(\alpha', \beta', \gamma', \delta')$, and as derived from the two planes (A, B, C, D), (A', B', C', D') is precisely the same in each case; and we may confine ourselves to the first point of view, regarding therefore the six coordinates as derived from the two points $(\alpha, \beta, \gamma, \delta)$, $(\alpha', \beta', \gamma', \delta')$. I further remark, that I do not at present in anywise fix the absolute magnitudes of the coordinates (a, b, c, f, g, h): it is only the ratios that we are concerned with.

8. The values of the ratios $a : b : c : f : g : h$ of the six coordinates do however depend on the particular coordinate planes $x = 0$, $y = 0$, $z = 0$, $w = 0$, made use of for their determination; and in the sequel it will be necessary to investigate the formulae of transformation to a new set of coordinate planes $x_0 = 0$, $y_0 = 0$, $z_0 = 0$, $w_0 = 0$. And I shall also show in what manner the absolute magnitudes of the coordinates may be fixed. But deferring the consideration of these questions, I consider the planes $x = 0$, $y = 0$, $z = 0$, $w = 0$ as given planes, and take the six coordinates (a, b, c, f, g, h) of a line to be determined as above in reference to these given planes, the absolute values of these coordinates remaining indeterminate, and their ratios only being attended to. And I proceed to consider the various questions which present themselves in the geometry of the line, considered as thus determined by means of its six coordinates (a, b, c, f, g, h).

Article, Nos. 9 to 18. (Various Sub-headings.) Elementary Theorems.

Condition that a line may be in a given plane.

9. Taking the line to be (a, b, c, f, g, h), the equation of the given plane to be

$$(A, B, C, D \between x, y, z, w) = 0;$$

then if $(\alpha, \beta, \gamma, \delta)$, $(\alpha', \beta', \gamma', \delta')$ are the coordinates of any two points on the line, we have the system of equations *ante*, No. 2, and substituting therein for $\beta\gamma' - \beta'\gamma$, &c. the values (a, b, c, f, g, h), we find

$$\begin{vmatrix} 0 & c & -b & f \\ -c & 0 & a & g \\ b & -a & 0 & h \\ -f & -g & -h & 0 \end{vmatrix} (A, B, C, D) = 0;$$

which equations, equivalent to a twofold relation, are the required condition. It may be remarked that, treating (A, B, C, D) as current plane coordinates, each equation of the system is that of a point lying in the line.

Condition that a line may pass through a given point.

10. The coordinates of the given point are taken to be $(\alpha, \beta, \gamma, \delta)$. If

$$(A, B, C, D \between x, y, z, w) = 0, \qquad (A', B', C', D' \between x, y, z, w) = 0,$$

are the equations of any two planes through the line, then we have the system of equations *ante* No. 3, and substituting therein for $AB' - A'B$, &c. their values in terms of the coordinates (a, b, c, f, g, h) of the line, we have

$$\begin{vmatrix} 0 & h & -g & a \\ -h & 0 & f & b \\ g & -f & 0 & c \\ -a & -b & -c & 0 \end{vmatrix} (\alpha, \beta, \gamma, \delta) = 0;$$

which equations, equivalent to a twofold relation, are the required condition. It is obvious that, treating $(\alpha, \beta, \gamma, \delta)$ as current point coordinates, each equation of the system is the equation of a plane through the given line.

Condition for the intersection of two lines.

11. The coordinates of the lines are taken to be (a, b, c, f, g, h), and $(a_1, b_1, c_1, f_1, g_1, h_1)$, respectively. If $(\alpha, \beta, \gamma, \delta)$, $(\alpha', \beta', \gamma', \delta')$, are the coordinates of any two points in the first line, and $(\alpha_1, \beta_1, \gamma_1, \delta_1)$, $(\alpha_1', \beta_1', \gamma_1', \delta_1')$, are the coordinates of any two points on the second line, then the four points are in a plane, that is, we have

$$\begin{vmatrix} \alpha & \beta & \gamma & \delta \\ \alpha' & \beta' & \gamma' & \delta' \\ \alpha_1 & \beta_1 & \gamma_1 & \delta_1 \\ \alpha_1' & \beta_1' & \gamma_1' & \delta_1' \end{vmatrix} = 0,$$

that is, expanding the determinant and substituting for $\beta\gamma' - \beta'\gamma$, &c. and $\beta_1\gamma_1' - \beta_1'\gamma_1$, &c. their values in terms of the coordinates of the two lines respectively, we have

$$af_1 + bg_1 + ch_1 + fa_1 + gb_1 + hc_1 = 0,$$

or, as this may also be written,

$$(f_1, g_1, h_1, a_1, b_1, c_1 \between a, b, c, f, g, h) = 0,$$

for the condition that the two lines may intersect.

12. The same result will be obtained if we take

$$(A, B, C, D \between x, y, z, w) = 0, \qquad (A', B', C', D' \between x, y, z, w) = 0,$$

for the equations of any two planes through the first line, and

$$(A_1, B_1, C_1, D_1 \between x, y, z, w) = 0, \qquad (A_1', B_1', C_1', D_1' \between x, y, z, w) = 0,$$

for the equations of any two planes through the second line. The four planes will meet in a point, that is, we have

$$\begin{vmatrix} A & B & C & D \\ A' & B' & C' & D' \\ A_1 & B_1 & C_1 & D_1 \\ A_1' & B_1' & C_1' & D_1' \end{vmatrix} = 0,$$

or, expanding and substituting, we have the same condition as before.

13. In the case of any two lines (a, b, c, f, g, h), and $(a_1, b_1, c_1, f_1, g_1, h_1)$, we may define the 'moment' of the two lines to be the function

$$af_1 + bg_1 + ch_1 + fa_1 + gb_1 + hc_1,$$

it being understood that we have not as yet any complete quantitative definition of the moment; this being so, we have, in what precedes, the theorem that the moment of two intersecting lines is $= 0$.

Plane through two intersecting lines.

14. Let $(A, B, C, D \between x, y, z, w) = 0$ be the equation of the plane through the two intersecting lines (a, b, c, f, g, h) and $(a_1, b_1, c_1, f_1, g_1, h_1)$. We have two systems of equations, as in No. 9, and comparing the corresponding equations of the two systems, we find in the first instance

$$\begin{aligned} A : B : C : D = \lambda &: bf_1 - b_1f : cf_1 - c_1f : -(bc_1 - b_1c) \\ &= ag_1 - a_1g : \mu : cg_1 - c_1g : -(ca_1 - c_1a) \\ &= ah_1 - a_1h : bh_1 - b_1h : \nu : -(ab_1 - a_1b) \\ &= gh_1 - g_1h : hf_1 - h_1f : fg_1 - f_1g : \rho, \end{aligned}$$

where λ, μ, ν, ρ, are in the first instance unknown; the different sets of ratios are of course identical in virtue of the relation

$$(f_1, g_1, h_1, a_1, b_1, c_1 \between a, b, c, f, g, h) = 0,$$

and comparing them we have equations which lead to the values of λ, μ, ν, ρ; and we thus obtain more completely.

$$\begin{aligned} A : B : C : D &= f_1a + b_1g + c_1h : bf_1 - b_1f : cf_1 - c_1f : -(bc_1 - b_1c) \\ &= ag_1 - a_1g : a_1f + g_1b + c_1h : cg_1 - c_1g : -(ca_1 - c_1a) \\ &= ah_1 - a_1h : bh_1 - b_1h : a_1f + b_1g + h_1c : -(ab_1 - a_1b) \\ &= gh_1 - g_1h : hf_1 - h_1f : fg_1 - f_1g : f_1a + g_1b + c_1h. \end{aligned}$$

15. It is in these equations easy to verify the identity of the different sets of values: we ought, for instance, to have

$$\frac{a_1f + bg + h_1c}{fg_1 - f_1g} = \frac{ab_1 - a_1b}{a_1f + bg_1 + ch_1};$$

that is,

$$(h_1c + a_1f + b_1g)(h_1c + af_1 + bg_1) + (ab_1 - a_1b)(fg_1 - f_1g) = 0,$$

and, observing that

$$\begin{aligned} (a_1f + b_1g)(af_1 + bg_1) + (ab_1 - a_1b)(fg_1 - f_1g) &= (af + bg)(a_1f_1 + b_1g_1) \\ &= ch \cdot c_1h_1, \end{aligned}$$

the left-hand side is

$$\begin{aligned} &= ch_1\,(ch_1 + af_1 + bg_1 + a_1f + b_1g + c_1h), \\ &= ch_1\,(af_1 + bg_1 + ch_1 + fa_1 + gb_1 + hc_1) = 0. \end{aligned}$$

Point on two intersecting lines.

16. Let $(\alpha, \beta, \gamma, \delta)$ be the coordinates of the point of intersection of the two intersecting lines (a, b, c, f, g, h) and $(a_1, b_1, c_1, f_1, g_1, h_1)$. We have two systems of equations such as in No. 10, and comparing the corresponding equations of the two systems, we find

$$\begin{aligned} \alpha : \beta : \gamma : \delta &= L : ag_1 - a_1g : ah_1 - a_1h : gh_1 - g_1h \\ &= bf_1 - b_1f : M : bh_1 - b_1h : hf_1 - h_1f \\ &= cf_1 - c_1f : cg_1 - c_1g : N : fg_1 - f_1g \\ &= -(bc_1 - b_1c) : -(ca_1 - c_1a) : -(ab_1 - a_1b) : P, \end{aligned}$$

where L, M, N, P, are in the first instance unknown; but, comparing the different sets of values, we have equations for finding the values of these quantities, and we thus obtain the more complete system

$$\begin{aligned} \alpha : \beta : \gamma : \delta &= f_1a + b_1g + c_1h : ag_1 - a_1g : ah_1 - a_1h : gh_1 - g_1h \\ &= bf_1 - b_1f : a_1f + g_1b + c_1h : bh_1 - b_1h : hf_1 - h_1f \\ &= cf_1 - c_1f : cg_1 - c_1g : a_1f + b_1g + h_1c : fg_1 - f_1g \\ &= -(bc_1 - b_1c) : -(ca_1 - c_1a) : -(ab_1 - a_1b) : f_1a + g_1b + h_1c, \end{aligned}$$

where it is to be observed that the right-hand side considered as a matrix is the transposed matrix of that which occurs in No. 13, in the formula for $A : B : C : D$. The verification of the identity of the different sets of values can of course be effected as in No. 15.

Expression for an arbitrary plane through a line.

17. The condition in order that the plane $(A, B, C, D \between x, y, z, w) = 0$, may pass through the line (a, b, c, f, g, h), is the twofold relation given, No. 9; it is satisfied by any one of the four systems

$$\begin{aligned} A : B : C : D &= 0 : h : -g : a, \\ \text{or } &= -h : 0 : f : b, \\ \text{or } &= g : -f : 0 : c, \\ \text{or } &= -a : -b : -c : 0; \end{aligned}$$

and consequently also by

$$\begin{aligned} A : B : C : D = &\ (0, -h, g, -a \between \xi, \eta, \zeta, \omega) \\ &: (h, 0, -f, -b \between \xi, \eta, \zeta, \omega) \\ &: (-g, f, 0, -c \between \xi, \eta, \zeta, \omega) \\ &: (a, b, c, 0 \between \xi, \eta, \zeta, \omega); \end{aligned}$$

or, what is the same thing, by

$$\begin{aligned} A : B : C : D = &\ (0, h, -g, a \between \xi, \eta, \zeta, \omega) \\ &: (-h, 0, f, b \between \xi, \eta, \zeta, \omega) \\ &: (g, -f, 0, c \between \xi, \eta, \zeta, \omega) \\ &: (-a, -b, -c, 0 \between \xi, \eta, \zeta, \omega), \end{aligned}$$

where $(\xi, \eta, \zeta, \omega)$ are arbitrary: there is, however, no loss of generality in putting any two of these quantities $= 0$.

Expression for an arbitrary point in a line.

18. The condition in order that the point $(\alpha, \beta, \gamma, \delta)$, may lie in the line (a, b, c, f, g, h), is the twofold relation given, No. 10; it is satisfied by any one of the four systems

$$\begin{aligned} \alpha : \beta : \gamma : \delta &= 0 : c : -b : f, \\ \text{or } &= -c : 0 : a : g, \\ \text{or } &= b : -a : 0 : h, \\ \text{or } &= -f : -g : -h : 0; \end{aligned}$$

and consequently, also by

$$\begin{aligned} \alpha : \beta : \gamma : \delta = &\ (0, -c, b, -f \between x, y, z, w) \\ &: (c, 0, -a, -g \between x, y, z, w) \\ &: (-b, a, 0, -h \between x, y, z, w) \\ &: (f, g, h, 0 \between x, y, z, w); \end{aligned}$$

or, what is the same thing, by

$$\begin{aligned} \alpha : \beta : \gamma : \delta = &\ (0, c, -b, f \between x, y, z, w) \\ &: (-c, 0, a, g \between x, y, z, w) \\ &: (b, -a, 0, h \between x, y, z, w) \\ &: (-f, -g, -h, 0 \between x, y, z, w), \end{aligned}$$

where (x, y, z, w) are arbitrary: there is, however, no loss of generality in putting two of these quantities $= 0$.

Article Nos. 19 to 25. Geometrical considerations in regard to three, four, five, and six lines.

Before proceeding further, I will establish certain geometrical notions in regard to three, four, five, and six lines. I use the term 'tractor' to denote a line which meets any given lines.

19. Three given lines have an infinity of tractors; viz. these are the generating lines of a hyperboloid having the three given lines for directrices.